

Proteomics with MSstats

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Introduction

MSstats is the standard Bioconductor package for statistical analysis of bottom-up mass-spectrometry proteomics. It ingests peptide-level intensity from any search engine (MaxQuant, Skyline, OpenMS, Spectronaut) and fits mixed-effects models with protein-level contrasts for label-free (LFQ), TMT, and SRM/PRM workflows.

Prerequisites

Peptide-level MS data; experimental-design matrix (condition, bioreplicate, run).

Theory

For each protein, MSstats fits a linear mixed-effects model with run/feature random effects and condition fixed effects; the contrast produces a log-fold change and a moderated t-test analogue. Missing-value handling uses accelerated failure-time imputation or censoring (AFT). LOG2 intensity is the response scale.

Assumptions

Peptide intensity is a valid surrogate for protein abundance; missingness is either completely at random (MCAR) or left-censored at a detection limit (MNAR).

R Implementation

```
library(MSstats)

# Simulated MSstats-format table: peptide intensity x run x condition
set.seed(2026)
proteins <- paste0("P", 1:20)
pep_per_prot <- 3
runs <- paste0("R", 1:6)
conditions <- rep(c("ctrl", "trt"), each = 3)

df <- expand.grid(Protein = proteins,
                 Feature = 1:pep_per_prot,
                 Run = runs,
                 stringsAsFactors = FALSE)
df$Condition <- conditions[match(df$Run, runs)]
df$BioReplicate <- df$Run
df$Intensity <- 2^(rnorm(nrow(df), mean = 20, sd = 1))

# Inject DE in first 5 proteins
trt_rows <- df$Condition == "trt" & df$Protein %in% proteins[1:5]
df$Intensity[trt_rows] <- df$Intensity[trt_rows] * 2
```

```

# Mock up the MSstats-required column names and minimum columns
msdf <- data.frame(
  ProteinName = df$Protein,
  PeptideSequence = paste0("PEP", df$Feature),
  PrecursorCharge = 2,
  FragmentIon = NA, ProductCharge = NA,
  IsotopeLabelType = "L",
  Condition = df$Condition,
  BioReplicate = df$BioReplicate,
  Run = df$Run,
  Intensity = df$Intensity
)

# MSstats workflow: processing + protein-level model + contrast
# processed <- dataProcess(msdf)
# comparison <- matrix(c(-1, 1), nrow = 1,
#                       dimnames = list("trt_vs_ctrl", c("ctrl", "trt")))
# results <- groupComparison(contrast.matrix = comparison, data = processed)
# head(results$ComparisonResult)
head(msdf, 3)

```

Output & Results

Per-protein log-fold change, standard error, p-value, and adjusted p-value; MSstats also outputs processed data diagnostic plots (profile plots, condition plots) for QC.

Interpretation

“MSstats identified 142 proteins differentially abundant at BH-FDR < 0.05 between treatment and control; the top hit (log2FC = 3.2) corresponds to the direct drug target.”

Practical Tips

- Provide replicate structure via `BioReplicate`; technical replicates must be distinct from biological replicates.
- Normalise within MSstats (`dataProcess(normalization = "equalizeMedians")`) rather than pre-normalising, to keep the pipeline single-entry.
- For TMT, use `MSstatsTMT`; the mixed-effects model differs due to isobaric design structure.
- Filter peptides to unique-to-protein before modelling; shared peptides inflate between-protein correlation.
- Report both LFC and adjusted p-value; very high LFC with wide CI (few peptides) is weak evidence.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)

- scRNA-seq with Seurat